Haeyong Chun, Ph.D.

Research Assistant Professor

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RESEARCH INTERESTS

My research advances and applies **psychometric methodology** to solve critical measurement challenges in health research. I specialize in factor analysis with ordered categorical indicators, measurement invariance, and Monte Carlo simulation to evaluate factor enumeration and scoring approaches. A complementary mission develops **computational psychometrics and physical-activity intervention models** leveraging AI—including LLM-based persona generation—to improve measurement quality and enable scalable, personalized behavior change support.

Core areas: Psychometric Methodology · Physical Activity Interventions · Monte Carlo Simulation · Structural Equation Modeling · LLM-Based Computational Psychometrics · Population Health Analytics

EDUCATION

2021–2026 **Michigan State University (MSU)**, East Lansing, MI
Ph.D., Kinesiology · Concentration: Psychosocial Aspects of Sport & Physical Activity
Advisor: Dr. Nicholas D. Myers

2025–2026 **Michigan State University (MSU)**
Sports Analytics Graduate Certificate

2018–2020 **Chungnam National University (CNU)**, South Korea
Master of Sport Science · Concentration: Sports and Exercise Psychology
Thesis: Development of Athlete's Emotional Regulation Questionnaire

2011–2018 **Chungnam National University**, South Korea — Dual Major
Bachelor of Mathematics Education; *Bachelor of Sport Science*

TECHNICAL SKILLS

Languages	R (primary), Python, SQL
Statistical Software	Mplus, SPSS, Microsoft Excel
Analytics & Modeling	Structural Equation Modeling, Factor Analysis, Monte Carlo Simulation, Multilevel Modeling, Longitudinal Analysis, LLM-Based Persona Generation & Simulation
LLM Tools	Ollama, Prompt Engineering, JSON Schema Enforcement
Data Engineering	pandas, numpy, openpyxl, ETL pipelines, ThreadPoolExecutor parallelism
Version Control	Git, GitHub
Visualization	ggplot2 (R), matplotlib/seaborn (Python), R Shiny (introductory)

PUBLICATIONS

Peer-Reviewed Publications (*Corresponding)

1. **Chun, H.**, Malete, L., Pacewicz, C. E., Lee, S., & Myers, N. D. (in press). Using Monte Carlo Methods for Benchmarking Factor Enumeration Criteria: An Invited Review and Related Application for Research in Exercise and Sport. *Research Quarterly for Exercise and Sport*.
2. Ai, S., Park, S., Yun, D., **Chun, H.*** (2026). The Moderating Role of Leisure-Time Physical Activity in the Associations of Sensory Processing Sensitivity with Mental Health and Self-Evaluative Outcomes Among University Students, *Korean Journal of Sport Psychology, 37*(3), 73–91. https://doi.org/10.14385/KSSP.37.3.73
3. Myers, N. D., **Chun, H.**, & Lee, S. (2026). Estimating Relationships Between Measures Derived from Responses to a Physical Activity Self-Efficacy Scale and Theoretically Relevant External Variables: A Monte Carlo Study. Measurement in Physical Education and Exercise Science, Advance online publication. https://doi:10.1080/1091367X.2026.2721483
4. Myers, N. D., **Chun, H.**, & Lee, S. (2026). Determining the Internal Structure of Responses to a Physical Activity Self-Efficacy (PASE) Scale Under a Latent Variable Approach: A Monte Carlo Study. *Measurement in Physical Education and Exercise Science*, 1–17. https://doi.org/10.1080/1091367X.2026.2649843
5. **Chun, H.**, Myers, N. D., Pacewicz, C. E., Malete, L., Prilleltensky, I., & Lee, S. (2025). Two-dimensional latent physical activity self-efficacy and its relationships with physical activity: A secondary data analysis. *Measurement in Physical Education and Exercise Science*, *29*(4), 351–366. https://doi.org/10.1080/1091367X.2025.2462661
6. Pacewicz, C. E., Hill, C.R., **Chun, H.**, & Myers, N. D. (2024). Confirmatory factor analysis in physical education and exercise science: A review with recommendations. *Measurement in Physical Education and Exercise Science*, *28*(2), 146–171. https://doi.org/10.1080/1091367X.2023.2270466
7. Myers, N. D., Pacewicz, C. E., Hill, C.R., & **Chun, H.** (2023). Factor analysis with ordered categorical indicators and measurement of self-efficacy in physical activity contexts. *Measurement in Physical Education and Exercise Science*, 27(4), 332-351. https://doi.org/10.1080/1091367X.2023.2186789
8. Lee, S., Myers, N. D., Bateman, A.G., **Chun, H.**, Prilleltensky, I., McMahon, A., & Brincks, A.M. (2023). Measuring Multidimensional Well-being Actions with the Newly Expanded I COPPE Actions Scale in Adults with Obesity. *Measurement in Physical Education and Exercise Science*, 27(3), 207–223. https://doi.org/10.1080/1091367X.2022.2153595
9. Myers, N. D., Lee, S., **Chun, H.**, & Zhu, W. (2023). Measurement in Physical Education and Exercise Science (MPEES): A Summary of MPEES-Related Activities in 2022. *Measurement in Physical Education and Exercise Science*, 27(2), 85–96.
10. Park, S., **Chun, H.**, Etnier, J.L., & Yun, D. (2023). Exploring the Mediating Role of Executive Function in the Relationship between Aerobic Fitness and Academic Achievement in Adolescents. *Brain Sciences*, 13(4), 614. https://doi.org/10.3390/brainsci13040614
11. Myers, N. D., Lee, S., **Chun, H.**, & Silverman, S. (2022). Measurement in Physical Education and Exercise Science: A Summary of MPEES-Related Activities in 2021. *Measurement in Physical Education and Exercise Science*, 26(3), 256–265.
12. **Chun, H.**, Yun, D., & Park, S. (2021). An Exploring Study on the Athlete's Emotional Regulation Variables. *Korean Journal of Sport Psychology*, 32(2), 21–44.

Manuscripts Under Review (*Corresponding)

1. Lee, Y. & **Chun, H.*** Functional Disability Profiles, Physical Activity, and Poor Mental Health Days Among U.S. Adults: A Disability-Inclusive Latent Class Analysis of the 2022–2024 Behavioral Risk Factor Surveillance System. [under review].

2. Lee, Y., **Chun, H.**, Kim, N., & Stubbs, B. Nonlinear Association Between Sedentary Time and Depression in the Post-COVID-19 United States: Evidence from NHANES 2021–2023. [under review].
3. Zimmerman, C. T., Myers, N. D., Driska, A. P., Pacewicz, C. E., Kim, J., **Chun, H.**, & Chung, K. Effects of a Pregame Speech Intervention on Coach Pregame Speech Efficacy and Coaching Efficacy: A Multiple Study Investigation. [under review].
4. Kim, H., Park, Y., & **Chun, H.*** Psychological Mechanisms of Within-Session Persistence in Digital Health Interventions: Evidence from Large-Scale Behavioral Logs. [under review].
5. Lee, S., Vu, S. L., Kim, J., Myers, N. D., Sarac, H., Cho, Y., & **Chun, H.**, Hardersen, J., & Groathouse, D. (2026). Efficacy of the Confident Moves online intervention to promote physical activity and whole person health: Protocol for a randomized controlled trial. *Trial*. [under review].
6. **Chun, H.** LLM-Generated Investor Personas as Controlled Agents for Studying Sports Sentiment and Stock Returns. [under review].
7. **Chun, H.** & Chung, K. Digital Physical Activity Interventions for Youth and Adults With Overweight or Obesity: A Scoping Review of Design, Personalization, and Evaluation. [under review].
8. Son, V., Myers, N. D., Feltz, D., & **Chun, H**. "Power of We": Effects of Motivational Self-Talk and Synchrony on Performance, Efficacy Beliefs, and Sense of Unity in Dyadic Exercise. [under review].
9. **Chun, H.**, Park, S., & Yun, D. Reproducibility of Factor Structures in Large Language Model-Generated Responses from Korean Athlete Personas [under review].

Manuscripts in Progress (*Corresponding)

1. Son, V., Myers, N. D., Feltz, D., & **Chun, H.*** How Behavioral Synchrony and Cultural Mindsets Shape Efficacy and Unity in Dyadic Exercise or The Effects of Behavioral Synchrony and Individualism-Collectivism on Efficacy Beliefs and Sense of Unity in Exercise Dyads. [Manuscript Writing, Data Analysis].
2. **Chun, H.** Acquiescence and Subscale Correlations in Sport and Exercise Psychology Scales: A Monte Carlo Study of Item-Keying Allocation and Correction. [Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing].

CONFERENCE PRESENTATIONS

Poster Presentations

1. **Chun, H.**, Hong, C., & Park, S. (2026, October). Reproducibility of Factor Structures in Large Language Model-Generated Responses from Korean Athlete Personas. *64th Annual Conference of the Korean Alliance for Health, Physical Education, Recreation, and Dance (KAHPERD),* Jeju, South Korea.
2. **Chun, H.**, Myers, N. D., Malete, L., Lee, S., & Pacewicz, C.E. (2026, March). Determining the Dimensionality of Physical Activity Self-Efficacy (PASE): A Monte Carlo Evaluation of Factor Enumeration with Ordinal-ESEM. *2026 AASP Midwest Sport and Exercise Psychology Symposium*, DeKalb, IL, USA.
3. **Chun, H.**, Myers, N. D., Pacewicz, C.E., Malete, L., Prilleltensky, I., & Lee, S. (2025, June). Two-dimensional latent physical activity self-efficacy and its relationships with physical activity: A secondary data analysis. *NASPSPA Annual Conference*, Lake Tahoe, NV, USA.
4. Myers, N. D., Pacewicz, C.E., Hill, C.R., & **Chun, H.** (2024, June). Factor analysis with ordered categorical indicators and measurement of self-efficacy in physical activity contexts. *NASPSPA Annual Conference*, New Orleans, LA, USA.
5. **Chun, H.**, Park, S., Yun, D., & Myers, N. D. (2024, June). Development and initial validation of the emotional regulation questionnaire in Korean athletes. *NASPSPA Annual Conference*, New Orleans, LA, USA.
6. Myers, N. D., Lee, S., Bateman, A.G., Wekesser, M., Prilleltensky, I., McMahon, A., Brincks, A.M., & **Chun, H.** (2023, June). Initial validity evidence for responses to the newly developed well-being actions self-efficacy scale. *NASPSPA Annual Conference*, Toronto, ON, Canada.

7. Park, S.Y., Yun, D.H., **Chun, H.**, & Etnier, J.L. (2019, June). Cognitive performance as a mediator of the relationship between aerobic fitness and academic achievement. *NASPSPA Annual Conference*, Baltimore, MD, USA.
8. Park, S.Y., **Chun, H.**, Yun, D.H., & Seo, G.R. (2018, June). Depression, Stress, Health-related Quality of Life and Occupational/Leisure Time Physical Activity: The Korean National Health and Nutrition Examination Survey (2014 and 2016). *Asian South-Pacific Association of Sport Psychology 8th International Congress*, Daegu, South Korea.

Oral Presentations

1. **Chun, H.**, Lee, S., Pacewicz, C. E., Malete, L., & Myers, N. D. (2026, June). *Determining the Dimensionality of Physical Activity Self-Efficacy: A Monte Carlo Simulation of Factor Enumeration.* 2026 Annual Conference of the Korean Society of Sport Psychology (KSSP), Cheonan, South Korea.
2. **Chun, H.**, Kim, H., & Park, Y. (2026, March). Psychological Mechanisms of Within-Session Persistence in Digital Health Interventions: Evidence from Large-Scale Behavioral Logs. *2026 AASP Eastern Canada Sport and Exercise Psychology Symposium*, Niagara Falls, ON, Canada.

RESEARCH EXPERIENCE

2026-present **Research Assistant Professor** · CNU, South Korea
Project: *Development and Evaluation of a Mechanism-Based, Personalized Just-in-Time Adaptive Intervention (JITAI) Coaching System for Physical Activity Maintenance in Adults with Overweight and Obesity*
PI: Dr. Haeyong Chun

2025–2026 **Research Assistant** · MSU, USA
Project: *United States Tennis Association — Physical Activity of Tennis Participation*
Responsibilities: Data collection, intervention delivery, and data analysis
PI: Drs. Karin Pfeiffer, Jennifer Roth & Sanghoon Kim

2022–2023 **Data Collector** · MSU, USA
Project: *CHAMP After-School Program*
Measured participants' physical ability and motor skills. PI: Dr. Karin Pfeiffer

2019–2020 **Data Collector** · CNU, South Korea
Project: *Sport Conditioning Program*
Participant recruitment, measuring physical ability and skills, writing reports. PI: Dr. Jong-Gil Lee

2018–2019 **Research Assistant** · CNU, South Korea
Project: *Sport Issues, Collective Emotions and Emotional Dynamics*
Collected fan sentiment data from SNS during international sport events; conducted structured interviews. PI: Dr. Jong-Gil Lee

2018 **Research Assistant** · CNU, South Korea
Surveyed participants in sport psychological counseling. PI: Dr. Dae-Hyun Yun

2016–2018 **Research Assistant** · CNU, South Korea
Project: *Macro-emotional Sociological Approach to 'Sport Emotions': SNS and Big Data*
Collected and analyzed fan emotional reactions from social media during Olympic, World Cup, and Asian Games. PI: Dr. Jong-Gil Lee

AWARDS & RESEARCH GRANTS

2026– **Principal Investigator** · National Research Foundation of Korea (NRF), ₩200,000,000 (~$135,000)

2031 Topic: Development and Evaluation of a Mechanism-Based, Personalized JITAI Coaching System for Physical Activity Maintenance in Adults with Overweight and Obesity

2025 **Dr. Kenneth Harding Graduate Student Endowed Scholarship** ($1,000) · MSU

2025 **Dissertation Completion Fellowship** ($7,000) · MSU

2024 **Summer Research Fellowship** ($7,000) · MSU

2023 **Janet A. Wessel Award in Adapted Physical Activity** · Kinesiology, MSU

2022 **Summer Research Development Fellowship** ($6,000) · MSU

2021 **Outstanding Graduate Thesis Presentation** · 2021 Winter Conference, KSSP

2019 **Research Scholarship** ($5,400) · CNU

2018 **Outstanding Trainee**, 2018 Sport Big-data Courses · Korea Sports Promotion Foundation (KSPO)

TEACHING EXPERIENCE

2026- **Instructor** · Statistics in Sport Psychology · CNU, South Korea
Designed and delivered full course on applied statistics for sport psychology students.

2021–2025 **Instructor** · Outdoor Soccer, Indoor Soccer, Volleyball · MSU
Designed curriculum, delivered lectures, managed grading for 3 sport activity courses.

2021–2025 **Teaching Assistant** · Latent Variable Modeling · MSU (Lecturer: Dr. Myers)

2021–2023 **Instructor** · Golf I & II · MSU

2022 **Guest Lecturer** · Sports and Exercise Psychology · CNU, South Korea
Topic: Motivation in Sports and Exercise Psychology

2020 **Instructor** · Statistics in Sport Psychology · CNU, South Korea
Designed and delivered full course on applied statistics for sport psychology students.

PROFESSIONAL SERVICE

2023–Present **Journal Reviewer** · Measurement in Physical Education and Exercise Science; Journal of Sport & Exercise Psychology; Journal of American College Health

2021–2023 **Editorial Assistant** · Measurement in Physical Education and Exercise Science

2023 **Assistant for Data Analysis** · National Academy of Kinesiology 2023 Review of Doctoral Programs

2022–2025 **COGS Representative** · Kinesiology Graduate Student Organization, MSU

2022–2023 **Member** · Welfare Committee, COGS, MSU

2025 **Registration Desk Volunteer** · 2025 NASPSPA Conference

2025 **Graduate Assistant** · Institute for the Study of Youth Sports, MSU

2021–2022 **Research Development Committee** · Kinesiology Graduate Student Organization, MSU

2021–Present **Member** · We Are Korean Educators Organization, MSU

CERTIFICATIONS & AFFILIATIONS

2018 **Teaching Certificate (Grade 2)** · Secondary School Teacher of Mathematics · Ministry of Education, South Korea

2017 **Sport Psychology Consultant (Level 3)** · KSSP

Professional Affiliations: North American Society for the Psychology of Sport and Physical Activity (NASPSPA) · Korean Society of Sport Psychology